QQ-B-675B
November 8, 1974
SUPERSEDING
Fed. Spec. QQ-B-675A
July 19, 1963

FEDERAL SPECIFICATION

BRONZE, ALUMINUM; INGOTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers aluminum bronze ingots as specified herein.

1.2 Classification. Aluminum bronze ingot shall be of one of the following copper alloy numbers: 952, 953, 954, 955, 956, 957, 958.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

QQ-C-390 - Copper Alloy Castings (Including Cast Bar).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices

FSC 9650

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indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

B 30 - Copper-Base Alloys in Ingot Form for Sand Casting
B 148 - Aluminum-Bronze Sand Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical composition. The material shall conform to the chemical requirements specified in table I.

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Table I. Chemical composition, percent.[a/]

Copper Alloy Number	Copper, min.[b/]	Lead, max.	Iron	Nickel	Aluminum	Manganese	Silicon
952	86.0[c/]	-	2.5-4.0	-	8.5- 9.5	-	-
953	86.0[c/]	-	0.8-1.5	-	9.0-11.0	-	-
954	83.0[d/]	-	3.0-5.0	2.5 max.	10.0-11.5	0.5 max.	-
955	78.0[d/]	-	3.0-5.0	3.0-5.5	10.0-11.5	3.5 max.	-
956	88.0[c/]	-	-	0.25 max.	6.0- 8.0	-	1.8-3.3
957	71.0[d/]	0.03	2.0-4.0	1.5-3.0	7.0- 8.5	11.0-14.0	0.10 max.
958	78.0[d/]	.02	3.0-5.0	4.0-5.5	8.5- 9.5	0.8-1.5	.05 max.

[a/] By agreement between manufacturer and purchaser, analysis may be required and limits established for elements not specified in the table.

[b/] In reporting chemical analyses obtained by the use of instruments such as spectrograph, x-ray, and atomic absorption, copper may be indicated as "remainder".

[c/] Total named elements - 99.0 percent minimum.

[d/] Total named elements - 99.5 percent minimum.

3.1.1 An analysis of each lot of ingots shall be furnished by the contractor showing percentages of the elements designated.

3.1.2 Analysis shall be made regularly only for the elements specifically stated in table I. If the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.2 Mechanical properties. When required (see 6.2), the ingots, when remelted and cast into tension test coupons, shall conform to the mechanical property requirements prescribed in QQ-C-390 or ASTM B 148, as applicable, for the same alloy, based upon the tension test procedures referenced therein.

3.3 Form. The aluminum bronze ingots shall be furnished in commercial standard forms or shapes.

3.4 Marking. Unless otherwise specified in the contract or order (see 6.2), each ingot shall bear, either stamped or cast in, the following information so that its identity will be clear at the point of delivery:

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Manufacturer's name or other identifying symbol.

Name of the material.

Alloy designation.

Heat or melt number.

3.5 Workmanship. The aluminum bronze ingots shall be of uniform quality, clean and free from dross, dirt, gassed or burnt metal, and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. The maximum weight of a lot shall depend on the method of production of the ingot.

4.2.1 When metal is all from a single melt or heat, the lot size shall be not over 20,000 pounds of ingots except as provided in 4.2.1.1.

4.2.1.1 The provisions of 4.2.1 may be modified to establish lot sizes greater than 20,000 pounds for metal that is all from a single melt or heat if the larger lot sizes are agreed upon by the manufacturer and procuring agency. In such a case, the first 5,000 pounds of the heat will be called the first sublot, the last 5,000 pounds the last sublot and the middle 5,000 pounds the middle sublot and, to determine uniformity, each sublot shall be tested separately. Each sublot shall separately meet the requirements for chemical analysis and mechanical tests.

4.2.2 When a lot is made up of different heats, the lot size shall be not over 5,000 pounds except as provided in 4.2.2.1.

4.2.2.1 The provisions of 4.2.2 may be modified to establish lot sizes not over 20,000 pounds if the larger size is agreed upon between the manufacturer and the procuring agency.

4.3 Sampling. At least one ingot shall be selected for tests from each 5,000 pounds in a lot not over 20,000 pounds and from each sublot of lots over 20,000 pounds.

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4.3.1 For chemical analysis. For each lot not over 20,000 pounds and each sublot of lots over 20,000 pounds, the drilling for analysis shall be taken from one ingot of the ingots selected for test. The drilling from each lot and each sublot shall be tested separately and shall conform to the requirements in table I. The drillings shall be taken from metal below the surface of the ingot and shall consist of not less than 5 ounces per sample.

4.3.2 For tension tests. When required, after the samples for chemical analysis have been taken, at least two test coupons shall be cast from the selected ingots from each lot not over 20,000 pounds and sublot of lots over 20,000 pounds. Not less than 100 pounds of ingot from the sample representing each lot not over 20,000 pounds and each sublot of lots over 20,000 pounds shall be selected for remelting to make these tests. The test coupons shall be cast in sand.

4.3.3 For visual examination. From each lot of material, samples shall be selected for visual examination in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.4 Visual examination. Each sample selected in accordance with 4.3.3 shall be visually examined to determine compliance with the requirements for identification marking (see 3.4) and workmanship (see 3.5).

4.5 Test procedures.

4.5.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, referee analysis shall be by method 111.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.3 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5, subject to the provisions of the section on "Rejection and retests" of Fed. Test Method Std. No. 151.

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5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2).

5.1.1 Level C. The products shall be separated by size, alloy, and form and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking (see 6.2 and 6.4).

5.2.1 Civil agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Copper alloy ingots are used for remelting into casting conforming to the corresponding copper alloy numbers of QQ-C-390.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Alloy required (see 1.2).
(c) When mechanical properties of castings are required and the applicable specification therefor.
(d) When special product marking is required (see 3.4).
(e) Special marking, when required (see 5.2).

6.3 The chemical composition and mechanical properties of the material covered by this specification are equivalent to the chemical composition and mechanical properties of the corresponding copper alloy numbers of ASTM B 30.

6.4 The requirements for product identification marking (see 3.8) and for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government agencies and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

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MILITARY CUSTODIANS:

Army - MR
Navy - OS
Air Force - 84

Review activities:

Army - None
Navy - SH
Air Force - None

Preparing activity:

Army - MR

CIVILIAN AGENCIES INTEREST:

GSA-FSS

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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